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EX:NO:15- Alpha-Beta Pruning Algorithm for Gaming

AIM

To write a Python program to implement the Alpha-Beta Pruning algorithm for a gaming decision tree and display the tree graphically.

ALGORITHM

1. Create a game tree with leaf node values.
2. Initialize Alpha as negative infinity and Beta as positive infinity.
3. Traverse the tree using Minimax.
4. Update Alpha for MAX nodes and Beta for MIN nodes.
5. Prune branches when $\text{Alpha} \geq \text{Beta}$.
6. Display the game tree and the optimal value.

CODE:

```
import matplotlib.pyplot as plt
```

```
# Game tree
```

```
tree = {  
    "A": ["B", "C"],  
    "B": ["D", "E"],  
    "C": ["F", "G"]  
}
```

```
values = {  
    "D": 3,  
    "E": 5,  
    "F": 2,  
    "G": 9  
}
```

```
def alpha_beta(node, alpha, beta, maximizing):

    if node in values:
        return values[node]

    if maximizing:
        best = float("-inf")

        for child in tree[node]:
            value = alpha_beta(child, alpha, beta, False)
            best = max(best, value)
            alpha = max(alpha, best)

            if alpha >= beta:
                break

        return best

    else:
        best = float("inf")

        for child in tree[node]:
            value = alpha_beta(child, alpha, beta, True)
            best = min(best, value)
            beta = min(beta, best)

            if alpha >= beta:
                break
```

```

        return best

# Find optimal value
result = alpha_beta("A", float("-inf"), float("inf"), True)

print("Alpha-Beta Pruning")
print("-----")
print("Optimal Value =", result)


# Graph positions
pos = {
    "A": (0, 3),
    "B": (-2, 2),
    "C": (2, 2),
    "D": (-3, 1),
    "E": (-1, 1),
    "F": (1, 1),
    "G": (3, 1)
}

plt.figure(figsize=(8, 5))

# Draw lines
for parent in tree:
    for child in tree[parent]:
        x1, y1 = pos[parent]
        x2, y2 = pos[child]

```

```
plt.plot([x1, x2], [y1, y2], 'k-')
```

```
# Draw nodes
```

```
for node in pos:
```

```
    x, y = pos[node]
```

```
    if node in values:
```

```
        text = node + "\n" + str(values[node])
```

```
    else:
```

```
        text = node
```

```
plt.scatter(x, y, s=1500)
```

```
plt.text(x, y, text, ha="center", va="center", fontsize=12)
```

```
plt.title("Alpha-Beta Pruning Game Tree")
```

```
plt.axis("off")
```

```
plt.show()
```

